

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing and Making Connections

Oct 27 – 29 | 4.4 A Closer Look at Testing • 4.5 Making Connections | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CO10) 2. Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CO10) 3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CO3) 4. Compare bootstrap and randomization distributions by their center and use, identifying each center for given data. (CO3) 5. Use a 95% confidence interval to reach a two-tailed test conclusion at $\alpha = 0.05$, and decide whether a confidence interval or a hypothesis test fits a research question. (CO9, CO10)	e-Text: 4.4 A Closer Look at Testing	MO1, MO2	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	MO1, MO2	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)	MO1, MO2
		Interactive Lecture: 4.4 A Closer Look at Testing	MO1, MO2			Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	MO3, MO4, MO5
		Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	MO1, MO2				
		e-Text: 4.5 Making Connections	MO3, MO4, MO5	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered	MO3, MO4, MO5	Practice Test 2 (Blackboard, by Sun Nov 1)	None
		Interactive Lecture: 4.5 Making Connections	MO3, MO4, MO5			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	None
		Slides: 4.5 Making Connections	MO3, MO4, MO5				
						Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4, 4.5	All